

1 Short questions

a. What is meant by RE, ...?

- 2p for explaining the concept (for example: "The timing of taxes does not matter for household consumption choices", or just "changes in the tax do not affect household consumption")
- 2p for explaining the mechanism (for example "Households understand that a lower tax today must be compensated by higher taxation tomorrow") 1p for providing some motivation for the yes/no question (for example "No. Households are myopic and want to spend additional income immediately", "No. Many households are borrowing constrained and can spend more if taxes fall", etc, or "Yes. A reasonable approximation is that many households are forward-looking and understand that they or their children will have to pay for today's tax cuts.")

b. PPP

- 3p for explaining the concept
- 2p for yes/no question. Example "No. Many goods, in particular services such as a hair cut, are difficult to trade internationally so there is no reason to expect the price of such goods to be equalized across countries." References to transportation costs, taxes and tariffs, pricing to market, etc are also relevant. Example indicating that PPP is relevant: "Yes, transportation costs for many goods have fallen and e-commerce facilitate price comparisons and international competition." A reference to factor mobility would also be relevant (e.g. the hairdresser moving to a high-income country).

c. Taylor

- 2p for stating the Fisher equation (in terms of either inflation or expected
- 3p for stating the Taylor principle and explaining the logic behind it (just that i must increase more than either π or π^e for r to increase

d. Work and consumption

- 2p for formulating the correct budget constraint
- 3p for solving and finding $H = 1/\alpha$ and $C = w/\alpha$

e. fiscal

- 2p for explaining the difference: active fiscal policy needs new decisions, automatic stabilizers are elements of fiscal policy that, without new decisions, become more generous/expansionary in a recession and less so in a boom

- 1p each for relevant examples: automatic stabilizers, examples: the income tax system (taxes automatically increase in a boom), unemployment insurance (more people receive UI benefits in a recession). Active measures: new public infrastructure investments in a recession, ...

2 The Solow model

- a.
- b. I think that it is enough to state that $\dot{y} = (n+\delta)k$ or a similar expression and then illustrate this graphically.
- c.
- d. The analysis for output per capita should be straightforward; getting that correct is worth 6p. For full credit here they also must note: that there is initially a fall in consumption when s increases (3p) and that what happens to consumption in the long run (i.e. new ss compared to old ss) depends on if the new savings rate is higher than the golden rule savings rate or not (3p) (it is enough that they just say this - they do not have to identify the golden rule analytically)

3 AS-AD

- a. AD-curve (3p): The AD curve combines the IS curve and a monetary-policy rule. Important assumptions are: (i) that investment is negatively related to the (real) interest rate, (ii) that monetary policy responds with a higher interest rate when inflation increases, and (iii) that prices (or inflation or inflation expectations) are sticky so that the change in the policy rate results in a change in the real interest rate.
AS-curve (1p): The Phillips curve with an assumption of adaptive (or backward-looking) inflation expectations.
- b. With the new inflation target, the MP rule changes so that the central bank will set a lower interest rate for any given inflation level. The AD curve therefore shifts out.
 - As a result of this, the output gap increases and so does inflation.
 - In the next period, inflation expectations have increased as a result of the higher inflation in the first period. The AS curve therefore shifts up. Inflation therefore increases further, but the output gap is now less positive (since the CB raises the interest rate when we get closer to the new inflation target).
 - The economy gradually moves towards the new SS as the PC continues to shift up, but the AD curve remains the same as after the initial shift.

I suggest awarding up to 6p if students identify the initial shift of the AD curve and the new SS. And the remaining 6p for identifying how and why the PC curve gradually shifts up after the first period.

- c. In (b.) the central bank needed to generate a positive output gap so that inflation and then inflation expectations would increase. This is not necessary if they can raise inflation expectations by just communicating the new target. The PC curve then directly jumps up so that it intersects the new AD curve in the point where the output gap is closed.

Award 2p if they just say, without any motivation, that the economy moves directly to the new ss.

4 Essay, economic growth

- For Malthus, they should note that land is a production factor in fixed supply, that population growth is endogenous and depends positively on income per capita, and that the decreasing MPL leads to lower income per capita if the population grows. (4p)
- For Solow, they should mention the importance of physical capital accumulation, investment (saving), and the decreasing MPK but linear depreciation of capital. (4p)
- For Romer, they should mention the importance of R&D, investment in R&D (or ideas) and the non-rivalry nature of ideas. (4p)
- 6p for explaining how population growth makes long-term income increases impossible in Malthus, how population growth has no effect on long-term growth rate (but a negative level effect) in Solow, and how population growth leads to higher growth in Romer.

Additional 2p for other things they bring up. And all these points are a suggestion – there may be alternative things that they mention that are worth credit.

5 Growth accounting

- a. They should note that $g_y = g_A + \alpha g_k + (1 - \alpha)g_h$. With the numbers given, the contribution from capital is then $(1/3)*3=1$ percentage point and the contribution from labor is $(2/3)*0.3=0.2$ percentage points. The remaining 0.8 percentage points come from TFP.
- b They can either do the growth accounting by deriving the expression for Y in terms of K/Y as in problem set 2, or as:

$$\begin{aligned} g_y &= g_A + \alpha g_k + (1 - \alpha)g_h \implies \\ g_y - \alpha g_y &= g_A + \alpha(g_k - g_y) + (1 - \alpha)g_h \implies \\ g_y &= \frac{1}{1 - \alpha}g_A + \frac{\alpha}{1 - \alpha}g_{ky} + g_h \end{aligned}$$

Plugging in values, note first that $g_{ky} = g_k - g_y = 1\%$. We then have the contribution from k/y as $(1/3)/(2/3)*1=0.5$ percentage points. The contribution from h is 0.3 percentage points. The remaining 1.2 percentage points are then explained by TFP.